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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-101

1. Topic (12 words max)

CropGuard: Edge AI Active Learning System for Real-Time Crop Disease Detection

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

One of the biggest risks to agricultural production is still crop diseases, especially for smallholder farmers in emerging nations like Sri Lanka. To detect crop illnesses, these farmers usually rely on eye inspection, peer advice, or postponed specialist consultations, which might result in inaccurate or delayed treatment. The issue is made worse by inconsistent internet connectivity, a lack of real-time diagnostic tools, and restricted access to agricultural officers. Because of this, early-stage infections can develop into serious outbreaks that result in large crop losses and financial hardship for farming communities.

This issue has a multifaceted influence. Reduced agricultural yields pose a social danger to farmer livelihoods and food security. Economically speaking, improper pesticide use and misdiagnosis raise production costs while decreasing revenue. Technically speaking, most of current AI-based solutions are computationally demanding, cloud-dependent, and inappropriate for remote locations with poor connectivity. In terms of the environment and ethics, improper or excessive chemical use adds to water contamination, soil deterioration, and health hazards for farmers and consumers. These problems show how urgently accessible, precise, and context-aware disease detection technologies are needed.

This study fits into the AgriTech industry vertical and is located within the Focus on Computing research cluster of Artificial Intelligence and Intelligent Systems, where it intersects with Edge Computing and Applied Machine Learning.

By utilizing Edge AI to provide real-time, on-device crop disease detection and active learning to adjust to regional disease differences and changing field conditions, the proposed research tackles the issue. Additionally, the incorporation of an expandable information retrieval method guarantees that farmers receive specific, localized treatment assistance instead of general recommendations.

6. Brief description of Existing Research and Systems (300 words max)

Existing crop disease monitoring systems primarily use cloud-based deep learning algorithms and smartphone. Plantix, Crop Doctor, and numerous AI-powered mobile apps enable farmers to upload photographs of crop leaves to centralized servers, where convolutional neural networks analyze the images and provide disease forecasts as well as basic treatment options. These systems have exhibited remarkable accuracy under controlled conditions, contributing to the widespread adoption of AI-powered agricultural support.

Several academic studies have focused on employing deep learning techniques such as CNNs, transfer learning, and ensemble models to classify crop diseases using big image datasets. Many studies show promising accuracy scores when training models on datasets such as PlantVillage. Some recent studies look into multimodal inputs and explainable AI to boost trust and usefulness. However, most of these systems presume continuous internet connectivity and rely significantly on cloud infrastructure for inference and updates.

Despite these developments, numerous limits persist. First, cloud-based technologies cause latency, recurring data costs, and usability concerns for farmers in rural or remote places with unstable network connectivity. Second, most existing models are static; once deployed, they cannot respond to new disease variations, regional disparities, or changing environmental conditions unless manually retrained with new information. Third, present systems are frequently monolithic, making it impossible to provide support for new crops or illnesses without retraining or redesigning the entire model. Finally, treatment recommendations are usually broad, rule-based, or inadequately localized, limiting their practical utility for smallholder farmers.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution, CropGuard, is an intelligent edge-based crop disease detection system that addresses the limitations of cloud-dependent agricultural AI through offline Edge AI operation, active learning for continuous improvement, modular scalability, localized advisory support, and regional adaptation. Unlike traditional systems, CropGuard deploys a TinyML-optimized deep learning model on a smartphone, enabling real-time offline inference without internet dependency. The active learning pipeline flags low-confidence predictions for expert review and integrates them into regular model updates, allowing the system to continuously adapt to new disease variations and field conditions.

Following disease identification, the system provides localized treatment recommendations via a RAG-based component that retrieves verified information from trusted Sri Lankan agricultural sources and generates advice in Sinhala and Tamil. To overcome the "one-size-fits-all" limitation, a Region-Adaptive AutoML Framework fine-tunes the base model with region-specific data from areas like Jaffna and Galle, automatically selecting optimal configurations to adapt to local climate and disease patterns, creating highly accurate models for different farming environments.

Input:

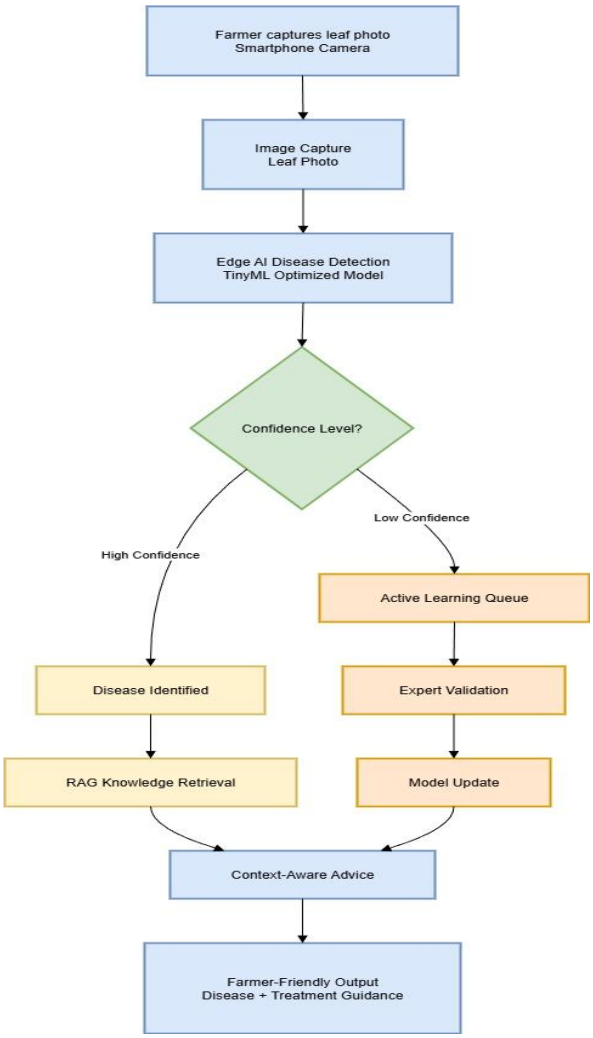
The primary input is an image of a crop leaf captured by a farmer using a smartphone application. Optional contextual inputs such as crop type, location, and farming preference (e.g., organic or chemical treatment) may also be provided.

Process:

A compressed convolutional neural network optimized by quantization and pruning algorithms processes the collected image on-device. The model classifies diseases in real time offline. The image is designated for the active learning loop and safely synchronized when connectivity is available if the prediction confidence falls below a predetermined level. A retrieval-augmented generation (RAG) component that queries a curated agricultural knowledge base to obtain pertinent treatment and management recommendations comes after disease identification.

Output:

The identified disease, confidence level, and context-aware therapy recommendations—including chemical and non-chemical choices, safety guidelines, and best practices—are generated by the system. To facilitate quick decision-making, the findings are shown in an easy-to-use interface for farmers.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful implementation of the proposed Edge-AI and Active Learning framework for real-time crop disease detection requires a combination of agriculture domain expertise, artificial intelligence knowledge and access to relevant data sources. The project team can feasibly obtain and develop these competencies through a combination of academic resources, expert collaboration and publicly available datasets.

Gaining specialized domain expertise:

Essential agricultural domain knowledge, such as common crop diseases, symptom patterns, treatment practices and seasonal variations, can be acquired through multiple channels. Team members can consult publicly available resources from the Sri Lankan Department of Agriculture, agricultural extension service publications and research articles focused on crop pathology in topical regions. In addition, informal interviews and consultations with agricultural officers, university lecturers in agriculture or experienced local farmers can help validate disease characteristics and treatment relevance. Reviewing existing agricultural AI systems and agronomy case studies will further strengthen domain understanding. This blended approach ensures that the system remains context-aware and aligned with real-world farming practices.

Gaining technical knowledge and skills:

The technical requirements of the project include machine learning, deep learning for image classification, Edge AI optimization, active learning pipelines and retrieval-augmented knowledge systems. These skills can be developed through coursework already covered in the degree program, such as machine learning, artificial intelligence and software engineering modules. Supplementary knowledge can be gained through online learning platforms (eg: Coursera, edX, TensorFlow documentation) and on-device inference. Practical experience will be gained by implementing prototypes using frameworks such as TensorFlow Lite, PyTorch Mobile and Android development tools. Research papers and open-source repositories related to Edge AI and active learning will further support skill acquisition.

Data collection and availability:

The project will primarily utilize publicly available and well-established datasets such as the PlantVillage dataset for initial model training and benchmarking. To improve localization and robustness, additional data may be collected through voluntary image submissions from farmers or controlled image collection in agricultural fields, subject to consent. Images collected during real-world usage with low-confidence predictions will be selectively added to the active learning pipeline after expert validation. This approach ensures continuous model improvement while minimizing data collection overhead. No personally identifiable information is required and images will be anonymized and securely stores.

Ethical clearance considerations:

This project involves minimal ethical risk, as it does not collect sensitive personal data or conduct invasive experiments. However, ethical clearance may be required if real-world data is collected from farmers, particularly to ensure informed consent, data privacy and responsible use of images. The project will follow institutional ethical guidelines by anonymizing data, obtaining consent where applicable and ensuring transparency regarding data usage. Therefor, ethical approval will be sought if human participants are involved during data collection or system evaluation.

9. Objectives and Novelty

Main Objective *To create a TinyML-optimized crop disease detection model that can perform accurate real-time inference directly on low-end smartphones without internet access.*			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22065094	**Train the base deep learning model for crop disease detection and implement active learning pipeline.**	• Collect and preprocess crop leaf image datasets (public datasets + locally collected data). • Train a baseline CNN/YOLO model for disease classification using a diverse dataset. • Design a confidence-based uncertainty detection mechanism. • Implement data collection and storage for uncertain samples. • Develop an expert validation workflow for labeling new data. • Retrain the base model periodically with newly labeled data.	**Implements a human-in-the-loop active learning framework** that allows the system to continuously learn and adapt to real-world disease variations, lighting, and background conditions. This directly overcomes the static nature of current agricultural AI models, reducing dataset bias and improving robustness over time
IT22100320	**Fine-tune the base model using region-specific data to create specialized models**	• Collect or synthesize region-specific data (e.g., Jaffna, Galle) for fine-tuning. • Apply transfer learning to fine-tune the base model with localized data from each region.	**Proposes a Region-Adaptive AutoML Framework** that leverages transfer learning and automated machine learning to fine-tune the base model into specialized versions for different

	for different agricultural regions via AutoML and transfer learning.	• Implement an AutoML pipeline to automate model architecture selection and hyperparameter tuning for each region. • Compare the performance of region-adapted models against the general base model.	regions. This novelty enables the system to automatically generate highly accurate, context-aware models tailored to specific regional climates and disease patterns without training from scratch, ensuring superior performance compared to a one-size-fits-all approach.
IT22130716	**Design and implement TinyML optimization for all models.**	• Apply model compression techniques such as quantization, pruning, and knowledge distillation on both the base and all regional models. • Evaluate model size (<50MB), inference speed (<3s), and accuracy (>85%) on low-end smartphones (<2GB RAM). • Deploy and test the compressed models on target mobile devices. • Ensure seamless model switching between base and regional versions.	**Introduces a TinyML-optimized deep learning model** specifically designed for low-end smartphones, enabling fully offline, real-time inference. This addresses the critical connectivity and hardware constraints that make existing cloud-dependent systems unusable for rural farmers. The compression is applied to ALL models (base + regional) to ensure efficient deployment.
IT22059840	**Develop a retrieval-augmented generation (RAG) system to provide contextual and localized treatment recommendations.**	• Curate agricultural knowledge-base content • Implement retrieval mechanism based on detected disease and context • Integrate generation logic for treatment guidance • Validate accuracy, safety, and relevance of recommendations	Introduces a **RAG-driven decision support layer** that delivers localized, context-aware, and safety-focused treatment advice, moving beyond generic rule-based recommendations found in current solutions.

10. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	x	No	

b) The proposed topic exhibits novelty.

Yes	x	No	

c) The student group can successfully execute the proposed project.

Yes	x	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	x	No	

e) Any other comments:

May have to limit to one crop and show that it can be extended to other crops with at least one additional crop.

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Dharshana	Kasthurirathna	
Co-Supervisor	Dr.	Sanvitha	Kasthuriarachchi	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				